

What DeFi Rates Should Be, Measured on Morpho

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Introduction

You deposit USDC on Morpho or Aave and earn 3% to 5%. Is that the right yield? In April 2026 Tom Dunleavy (@dunleavy89) answered the question with bond math [1]. He split the yield into a risk-free rate, an expected loss and a series of risk premia, added them up to a fair yield of 12.55%, and concluded that the market pays far too little. The replies argued over whether his premia counted the same losses twice. That post is what prompted this note. We try to put numbers on those risks with on-chain data, and to judge whether the yield depositors receive is reasonable.

This research follows the same decomposition, with two changes. The risk-free rate is the three-month bill rather than the ten-year yield, which matches a deposit that can be withdrawn at any time. And credit loss will be measured on-chain. All the data come from Morpho on Ethereum and Base. Morpho’s markets are isolated, so that bad debt in one market falls on the lenders of that market alone, so a loss can be attributed to a specific market and a specific collateral, which a shared pool cannot do.

Our results suggest the risks we could quantify are smaller than they are usually assumed to be. Credit loss on USDC lending runs at 7 bps a year on Ethereum and under 1 bp on Base; operational loss is about 11 bps. Together they come to less than 0.2 % points, against the 1.5 % points Dunleavy assumes for credit loss alone.

But losses do not arrive evenly. After the losses that actually occurred, depositors were left with almost nothing to compensate them for tail risk in 2026. The buffer was -42 bps on Ethereum and $+7$ bps on Base. Dunleavy’s sense that depositors are underpaid holds for the most recent year, but the reason is more likely an underpriced tail than a high average loss.

1 How to decompose a lending yield

We look at a depositor who lends USDC on Morpho. The depositor puts USDC into a vault, and the vault allocates it to underlying lending markets, where it is lent to borrowers who post crypto or stablecoin as their collateral. The vault’s curator decides which markets receive the money and how much, so the losses a depositor bears depend on the vault’s allocation. We treat Ethereum and Base separately, because lending on the two chains is structured very differently. On Base, most USDC borrowing sits in a single cbBTC/USDC market, while on Ethereum it is spread across many more collateral types, and vaults take on a wider range of risk.

Following Dunleavy, we break the yield a depositor should earn into parts:

$$\begin{aligned} \text{required yield} = & \text{risk-free rate} + \text{credit loss} + \text{operational loss} \\ & + \text{liquidity premium} + \text{tail risk and capital.} \end{aligned} \tag{1}$$

For comparison, Dunleavy’s fair yield is built as

$$\begin{aligned}
12.55\% &= \text{risk-free } 4.30\% + \text{expected loss } 1.50\% \\
&\quad + \text{oracle } 0.75\% + \text{governance } 1.00\% + \text{composability } 1.25\% \\
&\quad + \text{regulatory } 1.25\% + \text{depeg } 0.50\% \\
&\quad + \text{liquidity } 0.50\% + \text{model uncertainty } 1.50\%.
\end{aligned} \tag{2}$$

Compared with his list, we regroup the terms. His expected loss and four of his risk items, become credit loss and operational loss. The losses these risks cause are already in the loss data, so adding them again would count them twice. His premium for model uncertainty becomes the charge for tail risk and capital. We keep the liquidity premium as a separate term and do not quantify regulatory risk.

We measure what depositors actually earned from vault share prices. A depositor who puts USDC into a vault receives shares at the current share price. Interest earned by the vault raises the share price; the curator’s fee and any recognized losses the vault absorbs slow or reverse its rise. The change in the share price over a period is therefore the depositor’s return net of fees and recognized losses.

Each day we compute every vault’s share price return and average across vaults, weighting each vault by its size on the previous day. Compounding the daily averages and annualizing gives each chain’s realized yield.¹ This is the standard time-weighted measure of fund performance: it shows what a dollar kept in the vaults throughout would have earned.

Subtracting the quantifiable terms of equation (1) from realized vault yield gives the buffer: what remains to compensate depositors for tail risk. A positive buffer leaves some room; a buffer at or below zero means tail risk was not compensated.

2 Credit loss: what lenders actually lost

On Morpho every loan is over-collateralized, so a default happens in two steps. First the position is liquidated. This happens when the collateral price falls far enough that the loan exceeds the market’s liquidation loan-to-value (LLTV), the maximum ratio of debt to collateral value, for example 86%. A liquidator then repays part of the debt in exchange for the collateral plus a small bonus. Second, the liquidation may fail to clear the debt. Take a borrower who owes \$1,000 of USDC against collateral that has fallen to \$900. With a 5% bonus, the liquidator takes all \$900 of collateral and repays about \$857. The remaining \$143 is bad debt, and it is shared by everyone who lent in that market.

Dunleavy [1] lists three kinds of DeFi default: smart-contract exploits, oracle or governance attacks, and composability cascades. All three are operational failures, which we return to in Section 3. The routine credit event in over-collateralized lending is a fourth kind, a liquidation that does not clear the debt. It is also the one that can be measured directly on-chain.

We measure expected credit loss as

$$\text{Expected Loss} = \text{liquidation intensity} \times P(\text{bad debt} \mid \text{liquidation}) \times \text{LGD}. \tag{3}$$

Liquidation intensity is the share of borrowing that is liquidated in a year. $P(\text{bad debt} \mid \text{liquidation})$ is the share of liquidated debt that the liquidation fails to clear. LGD is the fraction of that uncleared debt that is finally lost. Multiplying the three terms gives the same number as

¹The sample covers 116 USDC vaults on Ethereum and 48 on Base. A vault is left out on any day when its size on the previous day was below \$1 million, and six vaults with implausible share prices, for example a constant gain of 2.2% a day, are excluded entirely.

dividing bad debt by borrowing. The point of splitting them is to see where a loss comes from: whether positions are liquidated often, whether liquidations fail often, or whether failures are severe.

The numerator comes from Morpho’s liquidation events on Dune, joined to the parameters of each market; we keep every market whose loan asset is USDC. The denominator is the daily borrowing in those markets, taken from Morpho’s API.² The window runs from April 2024 to September 2026.

Every term is measured in dollars rather than in number of events, because a depositor earns yield on each dollar lent and loses dollars, not events. The distinction is large (Table 1). On Ethereum, 9.2% of liquidations left bad debt, but those liquidations accounted for only 0.52% of liquidated debt, so counting events would overstate the rate 17.8 times. On Base the factor is 13.2. Most bad-debt liquidations are dust. Of Ethereum’s 206 events, one was a \$1.18 million loss; the other 205 add up to about \$6,000, and the median is about \$3.

Table 1 gives the inputs and results for the full period.

	Ethereum	Base
<i>Panel A: inputs</i>		
All liquidations		
Number	2,235	24,809
Debt involved	\$229.1M	\$485.4M
Liquidations that left bad debt		
Number	206	405
Debt involved	\$1.19M	\$0.60M
of which unpaid (bad debt)	\$1.18M	\$0.11M
Average USDC borrowing	\$695.7M	\$721.3M
<i>Panel B: results</i>		
Liquidation intensity (per year)	13.8%	28.2%
$P(\text{bad debt} \mid \text{liquidation})$, by amount	0.52%	0.12%
$P(\text{bad debt} \mid \text{liquidation})$, by count	9.2%	1.6%
LGD	0.998	0.187
PD (per year)	0.072%	0.035%
EL (per year)	7.1 bps	0.7 bps

Table 1. Credit loss in Morpho USDC markets, April 2024 to September 2026 (2.38 years). Debt involved in a liquidation is the debt the liquidator repaid plus any debt left unpaid. Liquidation intensity is debt involved in all liquidations divided by average borrowing per year. $P(\text{bad debt} \mid \text{liquidation})$ is debt involved in liquidations that left bad debt, divided by debt involved in all liquidations. LGD is bad debt divided by debt involved in liquidations that left bad debt.

Base liquidates positions twice as often as Ethereum. But a four times smaller share of its liquidated debt goes bad, and when it does, lenders lose about a fifth of it rather than nearly all of it. Expected credit loss comes to 7.1 bps a year on Ethereum and 0.7 bps on Base. Dunleavy assumes 150 bps, which is about twenty times Ethereum’s figure and more than two hundred times Base’s.

The losses did not arrive as a steady rate. By year, EL on Ethereum was 0.00, 0.02 and 18.7 bps in 2024, 2025 and 2026, and on Base 0.00, 1.9 and 0.1 bps. Almost all of Ethereum’s loss

²We exclude two frozen markets on Ethereum, PAXG/USDC and sdeUSD/USDC. Both have sat at 100% utilization since 2024 and 2025 respectively, where Morpho’s rate model charges its maximum rate; their recorded borrowing has compounded to several billion dollars of unpaid interest and does not reflect real lending.

comes from one liquidation in the RLP/USDC market, after the Resolv protocol was exploited in March 2026 [3]. Most of Base’s comes from one market whose collateral was an Aerodrome cUSDO/USDC LP token, after its price oracle was manipulated in May 2025 [4]. The largest USDC markets on both chains, cbBTC/USDC, recorded less than one cent of bad debt.

These figures include only bad debt that Morpho has recorded, and Morpho records it only when a position is liquidated. A market whose collateral collapses can freeze at 100% utilization and never produce that event. The largest case is sdeUSD/USDC on Ethereum. It was a listed market with about \$76 million of supply at its peak in May 2025. When Elixir’s deUSD collapsed in November 2025, the market froze with about \$9 million of debt and almost no collateral left, and Morpho delisted it [5]. If all of that is counted as loss, Ethereum’s EL rises to roughly 60 bps.

3 The other terms

Operational loss. For operational loss we draw on Bundi’s paper [2], which applies the Basel operational risk method to DeFi. He assembles about 1,075 hacks and exploits from February 2020 to May 2026, worth \$9.45 billion, and groups them by sector. For each sector he models how often losses occur and how large they are, with a heavy-tailed fit for the largest events, and expresses the expected annual loss as a share of the sector’s total value locked. Every protocol in a sector inherits the same rate, which gives Morpho 62 bps a year.³ His data also allow a narrower view. Filtered to Morpho itself, they contain two events: a misconfigured oracle in one market in October 2024 and an exploit of a peripheral contract in April 2025, together about \$2.8 million, or 2.1 bps a year. Two events are however too few to estimate a frequency. The true rate should lie between Morpho’s own record and the sector average since Morpho’s core contracts are small and cannot be upgraded, which argues for a rate below the sector’s, but two events cannot rule out luck. As an illustrative middle value, not an estimate, we use their geometric mean about 11.5 bps.

Liquidity premium. We set it to zero. A Morpho deposit can be withdrawn at any time, so it carries no term premium. What remains is the risk of not being able to withdraw: a market at 100% utilization cannot be exited until borrowers repay. On Morpho’s largest USDC market, cbBTC/USDC on Base, this lasted at most 11 hours at a time and has not happened since February 2025. Even at the upper end of a 95% confidence interval, it would affect deposits about 0.24% of the time.⁴ When a lock does occur, the rate rises sharply, and that extra interest is already in the realized returns we use in Section 4. Dunleavy cites Aave’s USDC market sitting at 99.87% utilization for four days after the Kelp exploit in April 2026. In the same week, Morpho’s cbBTC/USDC market peaked at 94.6% and did not lock.

Tail risk and capital. In bank loan pricing, the last term is a charge for capital. The lender holds capital against losses beyond the expected level, sized at a high quantile of the loss distribution (Basel uses 99.9%), and prices in the return that capital requires above the risk-free rate. Estimating this charge needs a credible tail for the loss distribution and a hurdle rate. Two and a half years of data, with only one or two large events per chain, cannot pin down a 99.9% quantile. We therefore do not estimate the term, and report the buffer defined in Section 1 instead. The cost of bearing tail risk cannot be negative, so a buffer at or below zero means the tail was not compensated.

³Bundi’s rates are a share of total value locked, while our other terms are a share of deposits or borrowing. The difference in denominators is small next to the uncertainty discussed here.

⁴No lock occurred from February 2025 to September 2026, about 1.6 years. With zero events, the upper end of a 95% confidence interval on the event rate is about three divided by the years observed, or about 1.9 locks a year; at the longest lock on record, 11 hours, that is at most 0.24% of the year.

4 Results

Table 2 brings our results together. The first rows are the parts of our formula that can be quantified; they add up to a required yield of 4.51% on Ethereum and 4.36% on Base. Below them is the yield that vault depositors actually earned: 5.96% on Ethereum and 4.19% on Base. The difference is the buffer. Figure 1 shows the same numbers next to Dunleavy’s 12.55%.

	Ethereum	Base	Dunleavy
Risk-free rate	4.32%	4.24%	4.30% ^a
Credit loss	0.07%	0.01%	1.50%
Operational loss	0.11%	0.11%	–
Other risk factors	incl.	incl.	3.50%
Regulatory risk	n.q.	n.q.	1.25%
Liquidity premium	0	0	0.50%
Tail risk and capital	see buffer	see buffer	1.50%
Required yield	4.51%	4.36%	12.55%
Realized vault yield	5.96%	4.19%	–
Buffer	+145 bps	–18 bps	–

Table 2. Required and realized yield on USDC deposits in Morpho vaults, per year. Ethereum covers April 2024 to September 2026; Base starts in June 2024, when its first USDC vault appeared, so the risk-free averages differ slightly. incl.: already included in credit and operational loss. n.q.: not quantified. ^a Ten-year Treasury; ours is the three-month bill averaged over the sample.

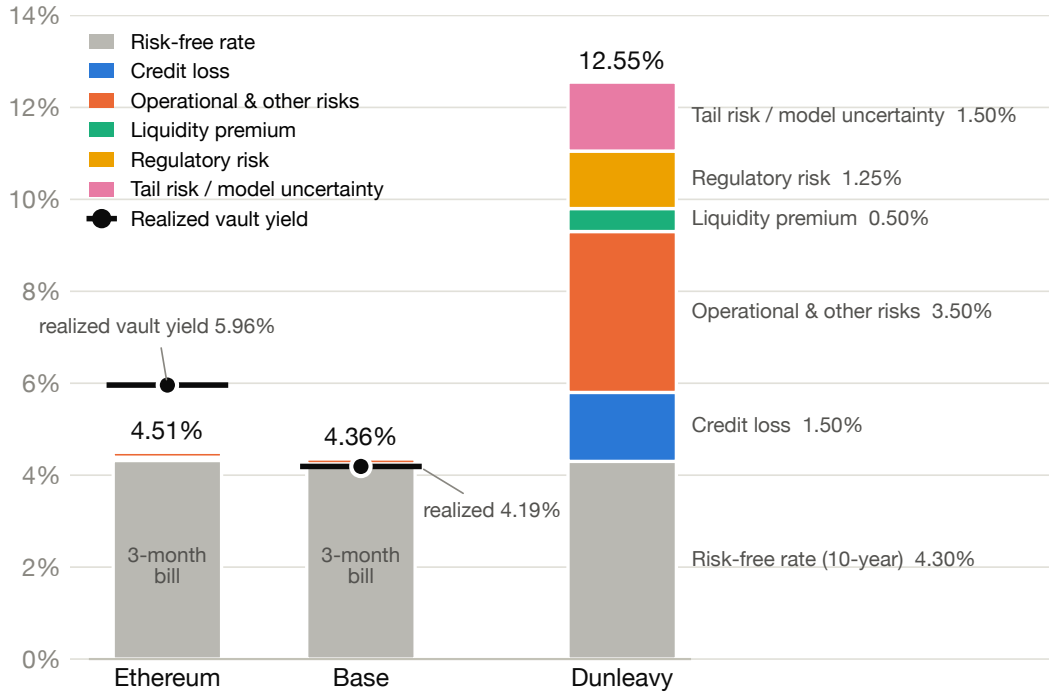


Figure 1. Required yield on Ethereum and Base, with realized vault yield marked, next to Dunleavy’s fair yield.

Compared with Dunleavy, our required yield is about 8 percentage points lower. The gap comes from three places. His credit loss assumption of 1.50% is more than twenty times what we

measure. He adds different types of risks worth 3.50% together. And he adds 0.50% for liquidity, 1.25% for regulatory risk and 1.50% for model uncertainty: we argue the first is zero, do not quantify the second, and treat the third as tail compensation. His risk-free rate is the ten-year yield rather than the three-month bill, but over our sample the two averages happen to be close.

Realized vault yield is measured from vault share prices (Section 1). On Ethereum it exceeds the required yield, leaving a buffer of +145 bps; on Base it falls short, leaving -18 bps. Because realized yield is already net of the losses that occurred, subtracting expected losses again makes the buffer slightly conservative, by no more than about 20 bps.

By year (Figure 2), the buffer on Ethereum was +311, +169 and -42 bps in 2024, 2025 and 2026; on Base it was -243 , +86 and +7 bps. Base’s vaults held only about \$40 million in 2024, when the chain was starting out. Since 2025 the buffer has narrowed on both chains, and on Ethereum it turned negative in 2026.

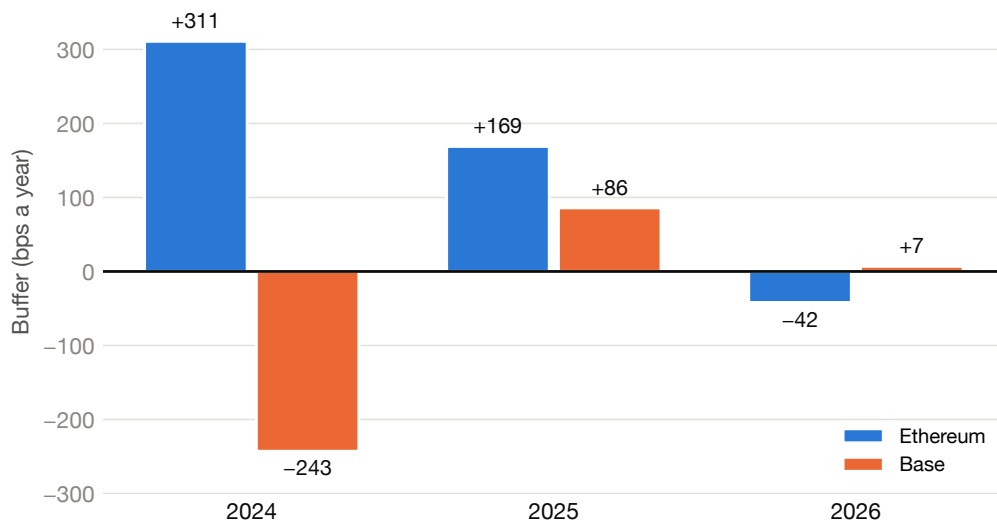


Figure 2. Buffer by year: realized vault yield minus required yield, in basis points. Base’s vaults held about \$40 million in 2024.

On Ethereum, the fall in the buffer lines up with two events. During the collapses of Stream Finance and Elixir’s deUSD in November 2025, Relend USDC lost 98.4% in a single day and Usual Boosted USDC lost 3.5%. After Resolv was exploited in March 2026, Gauntlet USDC Core lost 29.5% on 1 April. The unrecorded bad debt in sdeUSD/USDC from Section 2 shows up too: vaults that held that market have written the position off, which lowered their share price, so the loss is inside the realized yield. On Base, no vault in the sample lost more than 1% in a single day.

Where the buffer is zero or negative, no assumption is needed to conclude that tail risk went uncompensated. This covers Ethereum in 2026, on Base the buffer was close to zero. On Ethereum over the full period the buffer is positive, so depositors received some compensation for the tail, but we can not ensure whether it was enough for a hurdle rate.

The vaults on the two chains differ in risk profile. The Ethereum sample includes riskier vaults, while Base is mostly Prime vaults, so part of the gap between the chains reflects vault mix rather than the chain itself. Raising the vault size threshold from \$1 million to \$5 million or \$10 million changes the full-period buffer by less than 15 bps, because vaults are already weighted by size and small ones carry little weight.

5 Conclusion

Once each term is measured, the risks that can be quantified are smaller than they are usually assumed to be as Table 2 shows.

Although the risks together are small numbers, depositors were not paid enough to cover what is left. In 2026 the buffer was negative on Ethereum and close to zero on Base, so after the risk-free rate and the losses we can quantify, almost nothing remained to compensate for tail risk. Over two and a half years, the failures of Stream and Elixir’s deUSD, and then of Resolv, wiped out more than two years of accumulated interest at individual vaults in a single day.

These events are too rare on Morpho to fit a distribution. They still carry a practical implication for anyone about to deposit. With almost no margin left for the tail, a depositor has to watch what the vault holds, see which collateral the money is lent against, and decide whether that is a risk they want to take.

Data and code

Liquidation and bad-debt data come from Dune (<https://dune.com/queries/8715535/>). Market borrowing and vault share prices come from the Morpho API (<https://api.morpho.org/graphql>). The risk-free rate is the three-month Treasury yield from FRED (series DGS3MO). All queries and scripts are available at <https://github.com/badin-feio/morpho-usdc-yield-research>.

References

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